Agentic Crystallization v1.6

Stability Audit: A Trained Safety Margin for Cost-Aware Information Acquisition

Research status: Reproducible synthetic results. No phase transition or real-agent self-improvement established.

Abstract

Version 1.6 preserves all prior code and data, and tests whether a trained conservative margin improves the v1.5 Bayesian value-of-information (VOI) controller. The margin is selected on 360 training worlds, then frozen for two separate 720-world holdout cohorts. The selected margin (0.5 utility units) makes exactly the same decisions as the original zero-margin VOI controller in both cohorts. Added complexity provides no measured improvement. The v1.5 policy retains a context-dependent mean advantage over static Bayesian selection in the 75%-related environment but not the 25%-related environment. The evidence favors retaining the simpler controller, not asserting a stable crystallization threshold.

1. Hypothesis ledger

- H1: Nonadditive procedure interactions. Supported only within planted synthetic benchmark families.

- H2: Critical transition in adaptive efficiency. Not demonstrated.

- H3: Autonomous, persistent learning-to-learn. Not demonstrated.

- Engineering target: Better accepted utility per unit of computation under shifting task distributions. Not yet demonstrated in Pegasus.

2. Mathematical continuity

For disjoint procedure sets S and T, the higher-order finite-difference interaction is:

$$\Gamma_T(S)=\sum_{A\subseteq T}(-1)^{|T|-|A|}J(S\cup A).$$

The three-way AND task has a nonzero triple interaction at the empty baseline while its empty-baseline pairwise terms vanish. This is established Mobius algebra, not new mathematics. A candidate *engineering* potential is

$$F(S)=\alpha B(S)+\kappa V(S)-\beta U(S),$$

where B is boundary/coordination cost, V is verification cost, and U is independently measured future utility. It is not physical free energy. Structural graph connectivity and adaptive benefit are separate estimands.

3. Tested algorithm

Given current evidence h and remaining budget b, the model-estimated one-step value of a new cue C is:

$$\operatorname{VOI}(h)=\mathbb E_C[\max_\pi V_{b-1}(\pi\mid h,C)]-c_{probe}-\max_\pi V_b(\pi\mid h).$$

Version 1.6 tests the decision rule:

$$\operatorname{Probe}(h;m)=\mathbf 1\{\operatorname{VOI}(h)>m\}.$$

The margin m is selected from {0,0.1,0.5,1,2,4} on training-only data, maximizing mean utility with a conservative tie-break toward larger m. The selected m=0.5 is frozen before evaluation. No hidden target, latent relatedness, or second cue is passed to the controller before its decision. The cue consumes one of the same 40 attempts.

4. Experiment protocol

- Catalog sizes: 12, 15, 18. Related-task mixture: 25% or 75%.

- Model training seeds 3000–3399 (carried forward from v1.4).

- Margin training seeds 9000–9059: 360 worlds total.

- Holdout 1 seeds 10000–10119: 720 worlds total.

- Holdout 2 seeds 11000–11119: 720 worlds total, independent from holdout 1.

- Policies: no prior, static Bayesian, always probe, original zero-margin VOI, trained-margin VOI.

- Fixed exact-match oracle, paired worlds, 40-attempt budget, first validated procedure discovery cost charged to all policies.

- Utility = 40*found - 0.1*(first discovery attempts + second attempts including probes).

- No model inference, Pegasus runtime, live self-modification, or procedure promotion.

5. Results

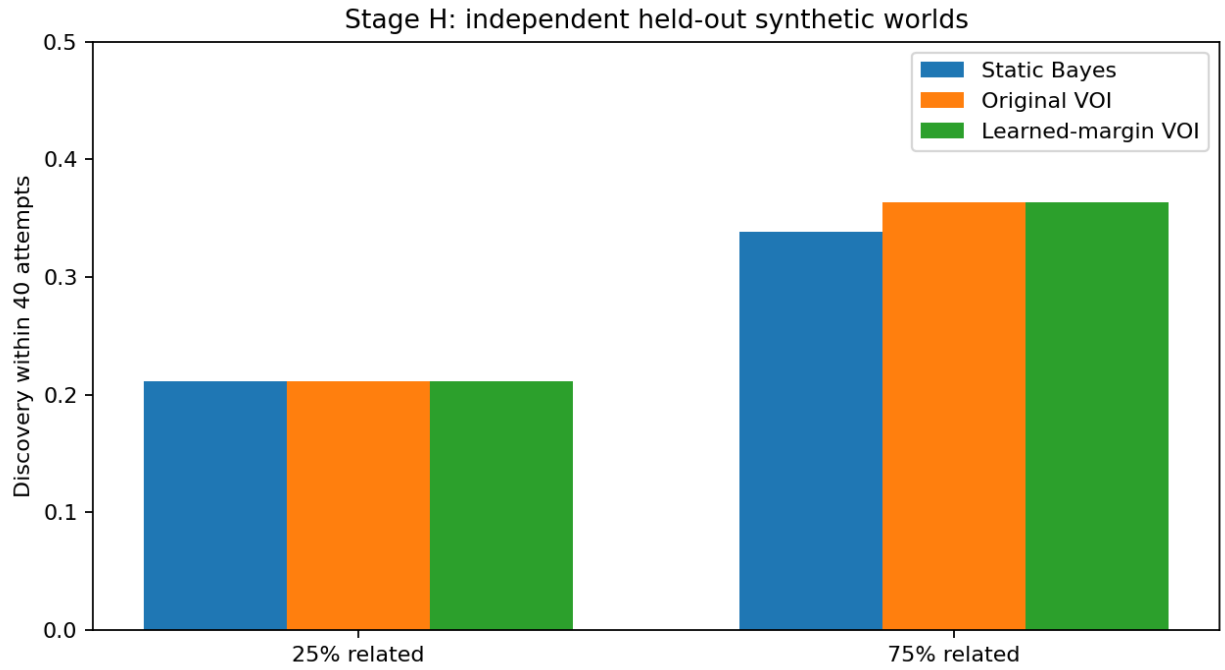
First holdout (360 worlds per mixture)

Relatedness	Static success	Original VOI success	Margin success	Margin minus static utility	Margin minus VOI utility
25%	21.11%	21.11%	21.11%	-0.016	0.000
75%	33.89%	36.39%	36.39%	+1.037	0.000

Independent replication (360 worlds per mixture)

Relatedness	Margin minus static utility	Margin minus original VOI	Margin minus always probe
25%	-0.120	0.000	-0.107
75%	+1.042	0.000	+0.239

Descriptive paired bootstrap 95% interval, first holdout, margin minus static: 25% related [-1.147,+1.114]; 75% related [-0.106,+2.241]. These are exploratory intervals, not preregistered confirmatory claims. The margin and original VOI have identical outcomes in both cohorts.



6. Scientific interpretation

The fitted margin fails to improve the original decision rule on these worlds. This is a negative model-selection result: retain the simpler algorithm. The VOI controller itself is not robustly validated under continuous distribution shift. A mean advantage under one synthetic mixture is not a coherence threshold, learning-to-learn, or evidence of a universal self-organizing principle.

A proposed product acceptance criterion is externally verified improvement over a strong static baseline on untouched tasks, no unacceptable negative transfer under specified shifts, bounded compute and latency, and preserved permissions and receipts. Those criteria are not met by these experiments.

7. Next research target

Before adding parameters, measure posterior calibration and probe-value estimation error under graded overlap and changing task distributions. Introduce a robust fallback only if it beats the simpler controller on independent holdouts. Independently test H2 using size-dependent causal hyperedge sweeps and finite-size scaling, with null models. Do not conflate smooth sigmoid fits with criticality.

8. Reproduce

```
PYTHONPATH=src OPENBLAS_NUM_THREADS=1 python -m unittest discover -s tests -v
PYTHONPATH=src python -m agentic_crystallization.stage_h --out experiments/reproduced_stage_h -
-start-seed 10000 --count 120
```

On Windows PowerShell, set $env:PYTHONPATH='src' and $env:OPENBLAS_NUM_THREADS='1' before running the Python commands.

[Source](../src/agentic_crystallization/stage_h.py) · [Tests](../tests/test_stage_h.py) · [Holdout CSV](../experiments/stage_h_holdout/runs.csv) · [Replication CSV](../experiments/stage_h_replication/runs.csv) · [v1.5 paper](archive/WHITE_PAPER_1.5.md).